THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture is formed when two or more substances are: 1

(a) mixed without chemical combination **(b)** converted to one element
(c) chemically combined in a fixed ratio **(d)** destroyed

2. Which is a uniform mixture? 1

(a) muddy water **(b)** sand and iron filings
(c) oil floating on water **(d)** sea water

3. Bronze is an alloy of: 1

(a) copper and tin **(b)** copper and zinc **(c)** lead and zinc **(d)** iron and nickel

4. A pure substance contains: 1

(a) different particles **(b)** many substances
(c) only air **(d)** the same type of particles

5. Which of these is a compound? 1

(a) Water **(b)** Sulfur **(c)** Hydrogen **(d)** Gold

6. Sodium chloride is made of sodium and chlorine in a ratio of: 1

(a) 1 : 2 **(b)** 2 : 1 **(c)** 3 : 1 **(d)** 1 : 1

7. Iron sulfide reacts with dilute hydrochloric acid to give iron chloride and: 1

(a) oxygen **(b)** hydrogen sulfide gas
(c) hydrogen only **(d)** carbon dioxide

8. Most rocks are: 1

(a) pure elements **(b)** liquids
(c) mixtures of minerals **(d)** gases

9. Magnesium burns in oxygen to form: 1

(a) magnesium chloride **(b)** magnesium oxide
(c) magnesium sulfide **(d)** magnesium only

10. Which is NOT a pure substance? 1

(a) Baking soda **(b)** Oxygen **(c)** Sugar **(d)** Fruit juice

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define a mixture and name its components' property. 2

12. Give two examples each of uniform and non-uniform mixtures. 2

13. What are alloys? Give two examples. 2

14. Define a pure substance. What are its two types? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain with an example why a mixture can be separated by physical methods. 3

16. Classify these as elements, compounds or mixtures: oxygen, sea water, carbon dioxide, gold, air, water. 3

17. Describe the activity in which iron and sulfur are heated. 3

OR

Explain how the properties of iron sulfide differ from iron and sulfur. 3

18. Explain the electrolysis of water as evidence that water is a compound. 3

OR

Describe the types of mixtures with examples of each combination (solid–solid, solid–liquid, gas–gas). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Define mixtures. Explain uniform and non-uniform mixtures, and alloys with three examples. 5

OR

Differentiate mixtures and compounds in a table of five points. 5

20. Explain pure substances and their two types with examples and the classification chart. 5

OR

Describe the activity of heating iron and sulfur. State observations, conclusions and the word equation. 5

END OF QUESTION PAPER — Total: 40 Marks

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